

Samuel Smith

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EDUCATION

Boston University	Boston, MA
Master's of Science, Mechanical Engineering	May 2028
<i>Coursework:</i> Electromechanical Design, Statics and Mechanics of Solids, Fluid Mechanics, Energy and Thermodynamics	
Ithaca College	Ithaca, NY
Bachelor of Science, Physics and Minor in Mathematics	May 2025
<i>Awards:</i> Departmental Honors in Physics, Dean's List 4 Semesters, Sigma Pi Sigma Honor Society	

EXPERIENCE

AI Trainer - CAD Expert	Remote
Handshake AI Fellowship	Dec 2025 - Apr 2026
- Developed and evaluated domain-specific prompts to assess performance of large language models (LLMs) in FreeCAD and Autodesk Inventor Professional - Engineered prompts for developing AI understanding of complex CAD designs through expert review and feedback	
Research Assistant	Ithaca, NY
Ithaca College	Jan 2025 - May 2025
- Built introductory 3D physics visualizations in Unity for use in undergraduate STEM instruction - Developed a reproducible workflow enabling educators to create custom 3D visualizations and presented project updates at two academic events	
Research Assistant	Ithaca, NY
Ithaca College	Jan 2024 - May 2024
- Redesigned five components of a Tolman-Stewart experiment apparatus in Fusion 360 - Led weekly progress reviews with research advisor to evaluate CAD redesigns, address design roadblocks, and outline next phase of apparatus development	

PROJECTS

Velocity Based Trainer (VBT) Product Design	May 2026 - Present
- Designed a VBT to measure barbell velocity using a spring-loaded spool and 600 PPR quadrature rotary encoder - Engineered ESP32-based firmware to process real-time encoder signals, including PCNT-based pulse counting, velocity calculation, and rep detection - Built a Python calibration and analysis pipeline validated against free-fall physics, achieving $\pm 2\%$ accuracy - Achieved a total device cost of \$100, below industry standard velocity based trainers priced above \$250	

SKILLS

Design: Solidworks, Prototype Design and Testing, Fusion360, Autodesk Inventor Professional
Software: Python (Matplotlib, NumPy, SciPy, Pandas) for Data Analysis
Embedded Systems: ESP32, Quadrature Encoder Signal Processing, Firmware Development (C++)

LEADERSHIP & AFFILIATIONS

Committee Director, International Model United Nations Association	Aug 2023 - Mar 2024
- Created an informative and reader-friendly report detailing international human rights challenges associated with weak online data protection policies - Mentored assistant directors to smoothly run a simulated UN committee of 250 student delegates - Provided individualized feedback on 60 student position papers	

EXTRACURRICULARS

Varsity Diving, Club Swimming, Consortium for Mathematics and Its Applications Competition 2022, 4 Semesters of Piano Performance Study